



A MA-SSA-optimized XGBoost-MLP framework using LIBS for rapid classification and quantitative analysis of heavy metals in traditional chinese medicines

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ABSTRACT

Laser-Induced Breakdown Spectroscopy (LIBS) holds significant value for rapid elemental detection; however, strong spectral interference, matrix effects, and high-dimensional data characteristics pose considerable challenges to accurate quantitative analysis. To enhance the performance of LIBS quantitative analysis, this study proposes a novel machine learning framework that integrates XGBoost and Multilayer Perceptron (MLP), optimized by a Multi-dimensional Adaptive Sparrow Search Algorithm (MA-SSA). The framework employs XGBoost for automated feature selection, eliminating redundant spectral variables while retaining critical information, and utilizes MA-SSA to optimize the hyperparameters of the MLP in regression tasks, significantly improving model stability and prediction accuracy. Experimental results demonstrate that the proposed method achieves 100 % accuracy in multi-class classification, outperforming traditional classifiers such as Random Forest, XGBoost, and standalone MLP. In terms of quantitative detection, the MA-SSA-optimized model achieves an RMSE of 4.43 $\mu\text{g/g}$, surpassing other hybrid optimization models including XGBoost-SSA-MLP (RMSE=4.62 $\mu\text{g/g}$), XGBoost-PSO-MLP (RMSE=5.225 $\mu\text{g/g}$), and XGBoost-GA-MLP (RMSE=5.584 $\mu\text{g/g}$). XGBoost-based feature selection effectively reduces spectral dimensionality while maintaining predictive performance. The proposed MA-SSA algorithm further enhances convergence efficiency and generalization capability. This study provides a robust, efficient, and scalable solution for LIBS analysis, with broad application potential in the field of real-time quantitative detection.

1. Introduction

Traditional Chinese medicinal materials (TCMM) are an important component of traditional medicine in the world and a treasure of Chinese medicine. TCMM have demonstrated good pharmacological activities, excellent curative effects, and extensive applications in China and globally [1]. According to a survey by the World Health Organization (WHO), about 60 %–80 % of the world's population uses TCMM [2]. However, many contaminants, especially heavy metals and pesticides, are increasingly detected in medicinal materials due to environmental

pollution, including soil and water, and during processing, storage, and transportation [3,4]. Heavy metal pollution is a severe problem, owing to its environmental ubiquity, non-biodegradability, persistence, and toxicity—even at trace levels—posing significant health risks including carcinogenicity through bioaccumulation in human tissues [5,6]. Therefore, the Chinese Pharmacopoeia (2025 version) issued by the National Pharmacopoeia Commission clearly stipulates the limit standards for heavy metals and harmful elements in TCMM. It can be seen that for the safety of the people, it is necessary to detect and research heavy metals in Chinese medicine.

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Current mainstream analytical techniques for heavy metal detection in traditional Chinese medicinal materials (TCMM) primarily encompass high-performance liquid chromatography (HPLC) [7], inductively coupled plasma optical emission spectroscopy (ICP-OES) [8], and inductively coupled plasma mass spectrometry (ICP-MS) [9]. Although these methods exhibit high detection sensitivity, they often necessitate intricate sample pretreatment and extended analysis times, rendering them unsuitable for field and rapid testing applications [10]. Consequently, developing a method for the accurate and rapid detection of heavy metal in TCMM is necessary.

Laser-induced breakdown spectroscopy (LIBS) is an analytical method with many advantages such as rapid and precise response, simultaneous multi-element analysis, high sensitivity [11]. Because of its advantages such as simple and in-situ detection, LIBS has been widely used in industry, environment and agriculture [12–15]. In recent years, machine learning has been widely applied in classification and regression prediction for spectral analysis because of its excellent feature-learning capability, which improves the accuracy of rapid spectral detection [16]. For example, Chen et al. used machine learning combined with LIBS to quantitative analysis of four metal elements in a single micro-carbon black particle. The results demonstrate that the RF model reduced the MRE, RSD, and RMSE from 0.1854, 363.7, and 434.7 ppm to 0.0866, 179.8, and 216.2 ppm, respectively [17]. Zhao et al. [18] established a CNN-assisted LIBS analytical method for brand classification of iron ores, and the accuracies of calibration and prediction reached 99.86 % and 99.88 %, respectively. However, LIBS data are often influenced by matrix effects and signal instability, making the selection of appropriate feature engineering and machine learning methods particularly critical for handling high-dimensional spectral data [19,20].

Within this context, Extreme Gradient Boosting (XGBoost) serves not only as a high-accuracy classifier but also as an effective tool for feature selection. Meanwhile, the Multi-layer Perceptron (MLP), leveraging its strong nonlinear mapping capability, is well-suited for modeling complex relationships in spectral data. Nevertheless, the performance of MLP is highly dependent on the selection of hyperparameters such as the number of hidden layers, neurons per layer, and activation functions. To address this optimization challenge, the Sparrow Search Algorithm (SSA) has been introduced, benefiting from efficient global search and rapid convergence. Despite these advantages, SSA exhibits inherent limitations in population initialization and update strategies, resulting in insufficient stability and generalization in high-dimensional optimization tasks. These issues often lead to premature convergence and inconsistent outcomes across repeated runs, ultimately compromising the reliability of the optimized MLP model. To overcome these drawbacks, this study proposes an enhanced technique named the Multi-dimensional Adaptive Sparrow Search Algorithm (MA-SSA). By systematically refining the population update mechanism and search strategy, MA-SSA significantly improves optimization stability and result reproducibility, offering a more robust solution for MLP hyperparameter tuning.

This study focuses on the classification and quantitative analysis of heavy metal concentrations in traditional Chinese medicines using LIBS. A multi-stage experimental framework was designed to validate methodological effectiveness and optimize model performance. Initially, an XGBoost-MLP model, which integrates XGBoost for feature selection with MLP for classification, was compared against standalone RF, MLP, and XGBoost models to demonstrate the improvement in classification accuracy attributable to XGBoost-based feature selection. Subsequently, for the regression task of predicting heavy metal concentrations, the XGBoost-MLP model was evaluated against PCA-MLP, PCA-RF, and PCA-SVM approaches, highlighting the advantage of feature selection over conventional PCA-based feature extraction. Finally, SSA, Genetic Algorithm (GA), and Particle Swarm Optimization (PSO) were employed to optimize the hyperparameters of the MLP component within the XGBoost-MLP model. While SSA yielded the best optimization

outcomes, its inherent instability prompted the development of an enhanced variant, called the MA-SSA. The proposed MA-SSA effectively addresses these limitations, resulting in an efficient and stable modeling framework suited for LIBS spectral analysis of traditional Chinese medicines.

2. Materials and methods

2.1. Sample preparation

The *Fritillaria*, *Ferula*, and *Cistanche* samples used in this study were purchased from a local pharmacy. Initially, the raw materials of each herb were ground into a fine powder using a crusher (AF-04A, Wenling Auli Traditional Chinese Medicine Machinery Co., Ltd., Wenling, China) and then sieved through a 200-mesh sieve to ensure particle size homogeneity. Subsequently, six portions of 3 g each were weighed from the powdered materials and transferred into beakers. These three sets of powder samples (18 portions in total) were intended for the subsequent machine learning classification task.

For the quantitative analysis, only the *Cistanche* samples were used to prepare the standard samples by spiking with copper (Cu). The intrinsic Cu concentration in *Cistanche* was determined to be extremely low and thus negligible compared to the spiked concentrations, ensuring no interference with the quantitative analysis. To prepare the calibration standards, appropriate volumes of a copper nitrate standard solution (Shanghai Aladdin Biochemical Technology Co., Ltd., Shanghai, China) were added to the *Cistanche* powder to achieve nominal Cu concentrations of 25, 50, 75, 100, 150, and 200 micrograms per gram ($\mu\text{g/g}$). Each mixture was thoroughly homogenized to ensure a uniform distribution of Cu. Finally, the spiked powders were compressed into tablets with a diameter of 30 mm and a thickness of approximately 5 mm using a powder tablet press (769YP-24B, Tianjin JingTuo Instrument Technology Co., Ltd., Tianjin, China) under a pressure of 25 MPa for 3 min.

2.2. Experimental setup

The LIBS experimental setup in our work is shown in Fig. 1. A nanosecond pulse laser (Nd:YAG Laser, Q-smart 850, Quantel Laser, Les Ulis, France) is used as the excitation light source, with a working wavelength of 1064 nm and a frequency of 10 Hz. Through a reflective mirror, the laser with a pulse energy of 150 mJ is focused onto the sample surface by a lens with a focal length of 100 mm. The sample is vertically clamped on a two-dimensional electric displacement platform (TC55V/DM525, iCAN Automation, Beijing, China). After the laser ablates the sample, plasma is generated and transmitted to a spectrometer (AvaSpec-ULS4096CL, Avantes B.V., Apeldoorn, the Netherlands) through a fiber optic probe. Its average spectral resolution is 0.1 nm, and the spectral range is from 200 to 832 nm. The nanosecond pulse laser and the spectrometer are controlled by a delay trigger (DG535, Stanford Research Systems, Sunnyvale, CA, USA). The spectrometer sends the collected data to a computer for display and analysis. In addition, the delay for the spectrometer is set as 1.5 μs for this measurement, and the integral time is about 1 ms.

3. Theoretical and computational methods

3.1. Multi-dimensional adaptive sparrow search optimization algorithm (MA-SSA)

The SSA is a novel heuristic optimization algorithm proposed by Xue et al. in 2020, inspired by the foraging and anti-predation behaviors of sparrows [21]. The algorithm balances exploration and exploitation by dynamically adjusting the search behavior of sparrows, where producers actively explore new areas while scroungers follow the best solutions found by producers. SSA has demonstrated superior performance in solving high-dimensional, nonlinear, and multimodal optimization

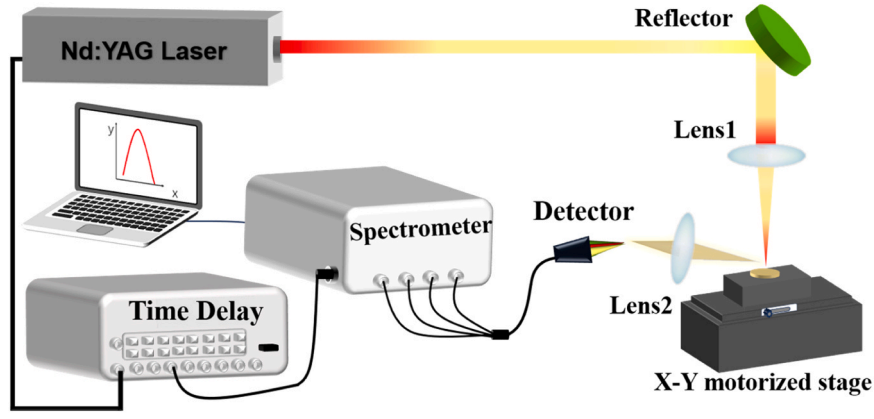


Fig. 1. The schematic diagram of LIBS experiment.

problems compared to traditional algorithms such as PSO and GA.

Despite its merits, the standard SSA exhibits certain limitations, particularly in terms of stability and generalization performance when applied to hyperparameter optimization for MLP models. To address these shortcomings, this study introduces a multi-dimensional enhanced version named the MA-SSA. The first key improvement aims at mitigating overfitting, which is a critical issue that arises when optimizing solely for training error. A generalization-weighted fitness function is proposed to explicitly incorporate generalization capability into the optimization objective. This function integrates the mean squared error on the training set (MSE_{train}), which reflects the fitting accuracy of hyperparameters, along with an overfitting penalty term based on the absolute difference between training and validation R-squared values ($|R^2_{train} - R^2_{val}|$). A larger discrepancy indicates more severe overfitting and thus incurs a stronger penalty. The enhanced fitness function is formulated as follows:

$$Fitness(x) = MSE_{train}(x) + \lambda \cdot |R^2_{train}(x) - R^2_{val}(x)| \quad (1)$$

where $MSE_{train}(x)$ denotes the mean squared error on the training set, $R^2_{train}(x)$ and $R^2_{val}(x)$ represent the coefficients of determination for the training and validation sets, respectively, and λ is the penalty coefficient that controls the strength of the overfitting penalty.

The second improvement introduces a diversity-adaptive producer step size mechanism to overcome the limitation of a fixed step size in the standard SSA. This approach dynamically adjusts the search step size based on real-time population diversity, effectively balancing global exploration and local exploitation throughout the optimization process.

The population diversity D is defined as the average variance across all hyperparameter dimensions. The formula is given by:

$$D = \frac{1}{n} \sum_{j=1}^n Var(x_j) \quad (2)$$

Where n represents the number of hyperparameter dimensions, and $Var(x_j)$ denotes the variance of the population positions in the j -th dimension. A higher value of D indicates greater diversity within the population.

The step size coefficient is adaptively adjusted using a threshold mechanism:

$$step_scale = \begin{cases} 0.1 & \text{if } D > \delta (\delta = 0.05) \\ 0.3 & \text{otherwise} \end{cases} \quad (3)$$

When diversity is high $D > \delta$, a smaller step size (0.1) is employed to facilitate localized fine-tuning. Conversely, when diversity is low, a larger step size (0.3) is used to encourage broader exploration.

To further enhance the search strategy, an exponential decay factor is incorporated into the position update rule for producers:

$$x_i^{(t+1)} = x_i^{(t)} \cdot \left(1 + N(0, 1) \cdot step_scale \cdot R_j \cdot \exp\left(-\frac{t}{\xi T}\right) \right) \quad (4)$$

In the producer position update formula: t denotes the current iteration index; $x_i^{(t)}$ represents the position of the i -th producer at iteration t ; $R_j = \max_j - \min_j$ indicates the range of the j -th dimension; T is the maximum number of iterations; $\xi \sim \text{Uniform}(0, 1)$ is a random coefficient; and $N(0, 1)$ denotes the standard normal distribution.

The final position is constrained within the feasible bounds using:

$$x_i^{(t+1)} = \text{clip}(x_i^{(t+1)}, \min_j, \max_j) \quad (5)$$

This adaptive mechanism significantly improves the algorithm's ability to maintain an optimal balance between exploration and exploitation across different stages of the optimization process.

The third enhancement addresses the issue of followers in the standard SSA overly relying on the global best solution, which often leads to premature convergence. A multi-guide collaboration strategy is proposed to improve population diversity through dynamically weighted guidance sources. This approach integrates information from multiple promising regions to guide follower updates, effectively balancing convergence and exploration.

The proposed multi-guide collaboration strategy incorporates three distinct guiding sources to mitigate premature convergence. The global best solution (G), defined as the individual with the lowest fitness value ($G = \arg\min_{x \in X} \text{fitness}(x)$), ensures convergence toward the current optimum. To maintain exploratory potential, the suboptimal individual (S), identified as the second-best solution in the fitness-sorted population, provides alternative directional information. Furthermore, the population center ($C = \frac{1}{N} \sum_{i=1}^N x_i$) serves as a stability anchor, preventing excessive clustering around specific regions.

A dynamic weight assignment mechanism is employed based on the follower's ranking within the population. For a follower with index i (where $n_p \leq i < N$, with n_p representing the number of producers and N the total population size), the normalized rank ratio $r = (i - n_p) / (N - n_p)$ (where $r \in [0, 1]$) is computed. This ratio inversely correlates with ranking quality—lower r values indicate higher-ranked individuals. The weight allocation is determined as follows:

$$\begin{cases} w_G = 0.5 + 0.4 \cdot r \\ w_S = 0.3 - 0.3 \cdot r, r = \frac{i - n_p}{N - n_p} \\ w_C = 0.2 \end{cases} \quad (6)$$

These weights are normalized to obtain the final weighting factors:

$$w'_k = \frac{w_k}{w_G + w_S + w_C}, k \in \{G, S, C\} \quad (7)$$

The position of the follower is then updated using:

$$x_i^{(t+1)} = w'_G \cdot G + w'_S \cdot S + w'_C \cdot C + 0.2 \cdot N(0, 1) \quad (8)$$

where $\text{fitness}(\cdot)$ denotes the fitness function; w'_G , w'_S , and w'_C represent the normalized weights; and $N(0, 1)$ denotes the standard normal distribution, which introduces stochastic directional perturbation.

3.2. Extreme gradient boosting algorithm (XGBoost)

XGBoost is an efficient and widely used machine learning algorithm based on the gradient boosting framework. Gradient boosting refers to a class of ensemble learning algorithms designed for classification and regression tasks. The ensemble is constructed through the sequential addition of decision trees, where each tree is trained to correct the residual errors of the previous ones. The core principle of XGBoost is the iterative construction of regression trees (CART) to progressively reduce prediction errors. The dataset $D = \{(x_i, y_i)\}$ consists of n samples, where $x_i \in \mathbb{R}^m$ represents the feature vectors and $y_i \in \mathbb{R}$ represents the target values. The predicted value at the k -th iteration is:

$$\hat{y}_i^{(K)} = \sum_{k=1}^K f_k(x_i), f_k \in \mathcal{F} \quad (9)$$

where K is the number of decision trees, f_k is the K -th submodel, x_i is the i -th input sample, and $\mathcal{F}$ is the set of all decision trees. The objective function of XGBoost consists of the loss function and the regularization term:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \gamma T_t + \frac{1}{2} \lambda \|w_t\|^2 \quad (10)$$

where t is the number of iterations, l is a differentiable loss function that measures the difference between the predicted value $\hat{y}_i$ and the target value y_i , $\hat{y}_i^{(t-1)}$ is the prediction from the previous iteration $t - 1$, γ and λ are regularization coefficients that prevent the decision tree from becoming overly complex, T is the number of leaf nodes, and w denotes the leaf weights.

In the XGBoost model, feature importance is assessed by measuring the contribution of each feature to the model's prediction performance. XGBoost uses several methods to calculate feature importance, with the most commonly used being Gain, Cover, and Frequency. In this analysis, we prioritize Gain as the metric for feature importance because it directly reflects the contribution of the feature to improving the model's predictive performance. For feature f_j , its importance can be represented by the average reduction in the loss function caused by all splits based on that feature. let S_j be the set of all splits based on feature f_j , and ΔL_s denote the loss reduction at split s . Then, the Gain importance $I_{\text{Gain}}(f_j)$ of feature f_j is defined as:

$$I_{\text{Gain}}(f_j) = \frac{1}{|S_j|} \sum_{s \in S_j} \Delta L_s \quad (11)$$

where $|S_j|$ is the number of splits based on feature f_j . In this way, Gain can quantify the contribution of the feature across the entire model. For ease of interpretation and comparison, we normalize the importance scores of each feature and convert them into percentage form. This makes the range of feature importance scores from 0 to 100, facilitating the intuitive comparison of the relative contributions of different features in the model. The normalization process can be completed using the following formula:

$$I_{\text{normalized}}(f_j) = \frac{I_{\text{Gain}}(f_j)}{\sum_{i=1}^N I_{\text{Gain}}(f_i)} \times 100\% \quad (12)$$

where $I_{\text{normalized}}(f_j)$ is the normalized importance of feature f_j , N is the total number of features, and $I_{\text{Gain}}(f_j)$ is the original gain importance of feature f_j . Through this normalization process, we can ensure that the sum of the importance scores of all features equals 100 %, thereby providing a clearer representation of the contribution ratio of each feature to the model's performance.

3.3. Multilayer perceptron algorithm (MLP)

The MLP [22] is a type of artificial neural network that consists of multiple layers of neurons, including an input layer, one or more hidden layers, and an output layer. The main goal of an MLP is to perform complex nonlinear transformations on data through multiple fully connected layers of neurons, establishing a mapping relationship between input features and output labels. It is commonly used for supervised learning tasks such as classification and regression. In regression tasks, the output layer typically uses a linear activation function, while the hidden layers use nonlinear activation functions to enhance the model's ability to perform nonlinear fitting. The output of the hidden layers can be expressed as:

$$h^{(l)} = \sigma(W^{(l)}h^{(l-1)} + b^{(l)}) \quad (13)$$

Where $h^{(l-1)}$ is the output of the $l-1$ layer, $\sigma(\cdot)$ is Nonlinear activation function, $W^{(l)}$ and $b^{(l)}$ are the weights and biases of the layer's neurons. In classification tasks, the activation function Softmax is typically used, and its expression is as follows:

$$P(\hat{y}_k | \mathbf{x}) = \frac{e^{\hat{y}_k}}{\sum_{j=1}^C e^{\hat{y}_j}} \quad (14)$$

The weight vector W and bias vector b are determined through the training process of the MLP, during which the MA-SSA optimization algorithm is used to optimize the weights and biases of the MLP model, thereby improving the prediction accuracy of the model in later stages.

3.4. MA-SSA optimization-based XGBoost-MLP model

First, we use the XGBoost algorithm for feature selection. XGBoost is a powerful Gradient Boosting Decision Tree (GBDT) algorithm with strong capabilities in feature selection. Through the feature importance evaluation mechanism of XGBoost, we can effectively identify the most critical features for model prediction. These features are then selected using a sequential backward selection (SBS) strategy, further optimizing the feature set. The selected important features serve as the basis for the input to subsequent models, while redundant or irrelevant features are removed, thereby reducing computational complexity and improving both the training efficiency and predictive performance of the model. As illustrated in Fig. 2, the XGBoost-MLP framework demonstrates the integration of feature selection and neural network training, where the output of XGBoost directly feeds into the MLP architecture.

Subsequently, we use the features selected by XGBoost as input data to train an MLP model. By inputting the XGBoost-selected features into the MLP model, we ensure that the training process is focused on the most relevant features for regression and classification tasks, thus enhancing the performance of both regression and classification models. To further enhance the performance of the MLP model, we introduced the MA-SSA algorithm to optimize its hyperparameters. In this study, the MA-SSA algorithm was employed to adjust the weights, biases, and other hyperparameters of the MLP, aiming to achieve optimal training results. Through this optimization process, MA-SSA can automatically adjust the

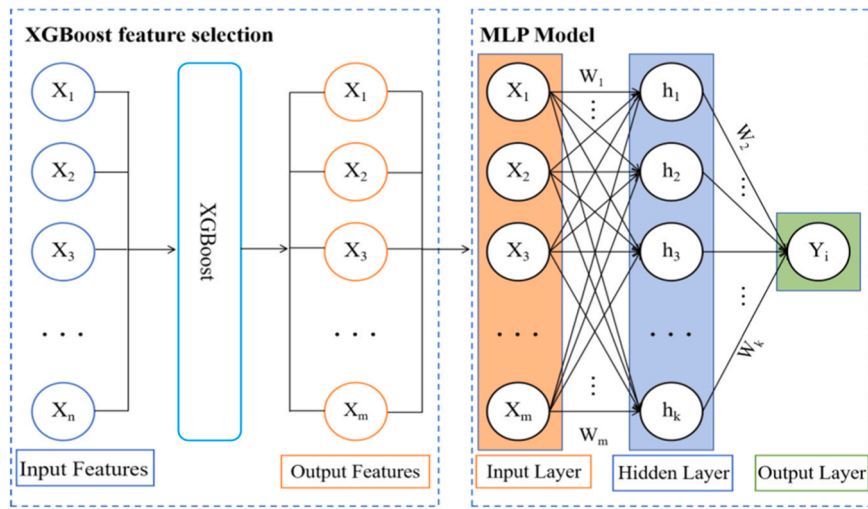


Fig. 2. XGBoost-MLP model structure diagram.

model's parameters during training, enabling the MLP to perform better on the test set and thus improving the overall performance of the model. The complete workflow of the hybrid MA-SSA-XGBoost-MLP model, including feature selection, MLP training, and hyperparameter optimization, is depicted in Fig. 3.

3.5. Model evaluation

In order to comprehensively evaluate the performance of our method across different tasks, we employed various metrics. For classification tasks, we used accuracy, precision, recall, and F1-score to assess the model's classification capability. Their calculation formulas are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (16)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (17)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (18)$$

where true positive TP is the number of samples that the model correctly predicts as positive; true negative TN is the number of samples that the model correctly predicts as negative; false positive FP is the number of negative samples that the model incorrectly predicts as positive; false negative FN is the number of positive samples that the model incorrectly predicts as negative.

For regression tasks, we utilized mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and the coef-

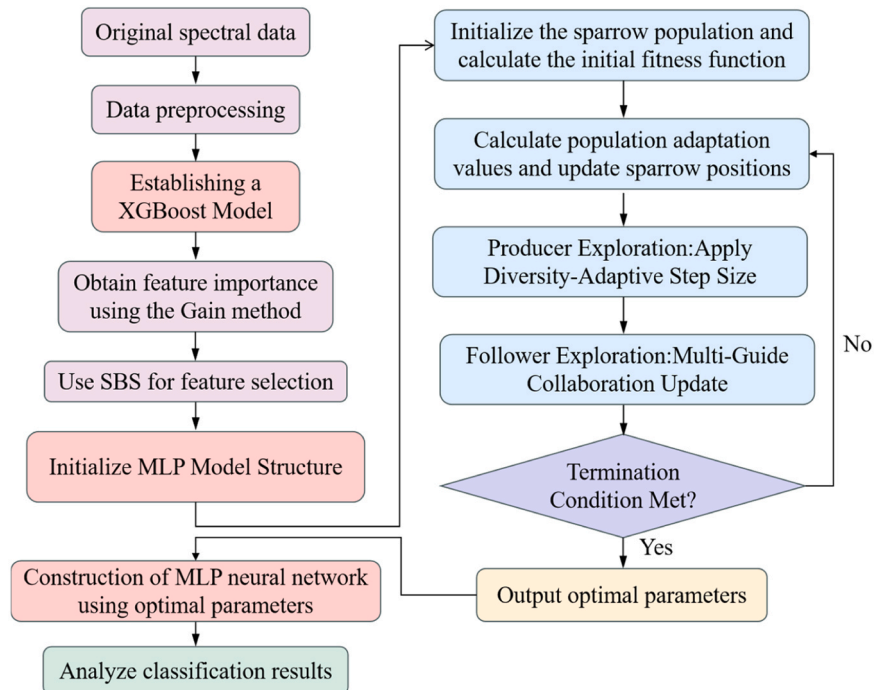


Fig. 3. Overall Process of MA-SSA-XGBoost-MLP model.

ficient of determination (R^2) to evaluate the model's prediction accuracy and fitting performance.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (19)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (20)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (21)$$

where n is the number of samples, y_i is the actual value, and $\hat{y}_i$ is the predicted value. These metrics allow for a thorough understanding of the model's performance across different tasks, ensuring its effectiveness and robustness in various scenarios.

4. Results and discussion

4.1. LIBS spectral analysis

This study systematically optimized a LIBS detection system to enhance sensitivity for heavy metal analysis. Critical improvements included refining laser focusing geometry and optimizing plasma emission collection optics. The characteristic copper emission lines at 324.75 nm and 327.39 nm were selected as analytical indicators for heavy metals. A delay time gradient experiment was conducted to determine optimal acquisition parameters. Fig. 4 illustrates the temporal evolution of spectral intensities for both copper lines under varying delay times. As shown in Fig. 4, the emission intensities of the copper lines decreased gradually with increasing delay time. Spectral stabilization occurred simultaneously at both 324.75 nm and 327.39 nm when the delay time reached 1.5 μ s, thereby establishing this parameter as the optimal detection condition.

Under the aforementioned optimized parameters, 200-laser-shot accumulative averaging was performed on *Fritillaria*, *Ferula*, and *Cistanche* samples to effectively mitigate plasma fluctuation effects on spectral data stability. This preprocessing protocol established a reliable data foundation for subsequent comparative studies of machine learning model performance. Fig. 5 displays characteristic spectral distributions of the three medicinal materials within the 220–800 nm range. Through alignment with the NIST atomic spectroscopy database, distinct emission lines of biologically essential elements were successfully identified, including Mg II (279.55/280.27 nm), Ca II (393.37/396.84 nm), and Na I (588.99/589.59 nm). The consistent detection of these elements validates the applicability of LIBS technology for compositional analysis in complex botanical matrices, while establishing a critical data benchmark for the development of advanced algorithmic models.

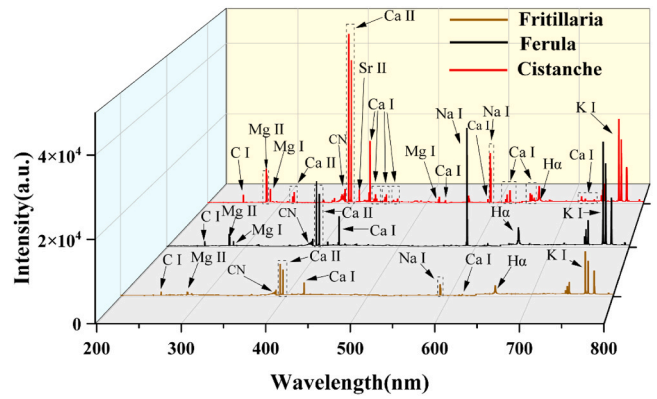


Fig. 5. LIBS spectra of *Fritillaria*, *Ferula*, and *Cistanche*.

4.2. Classification models

In this study, 100 sets of spectral data were collected for each of the three TCMM samples. From each dataset, 25 characteristic spectral lines were selected as input features for the model, generating a total dataset of 7500 data points (25 features multiplied by 300 samples). The dataset was partitioned into training and testing sets with a 7:3 split to ensure robust model evaluation. The dataset was subsequently fed into the XGBoost model to calculate the feature importance values. Fig. 6 illustrates the Gain-based feature importance of the top 8 most significant

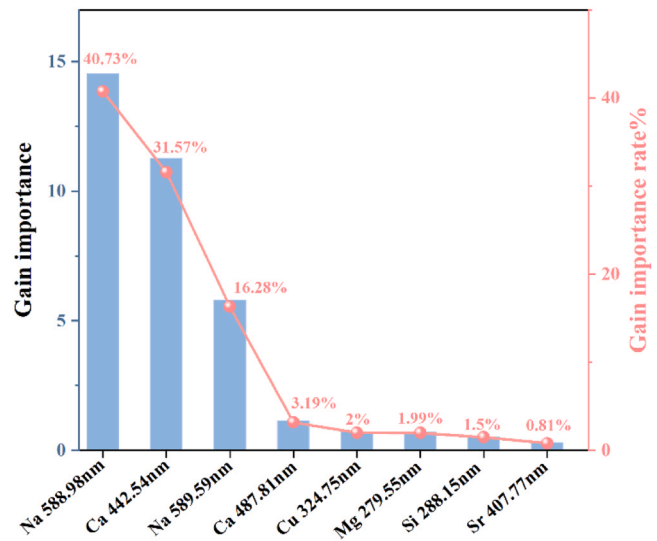


Fig. 6. Feature Importance Ranking.

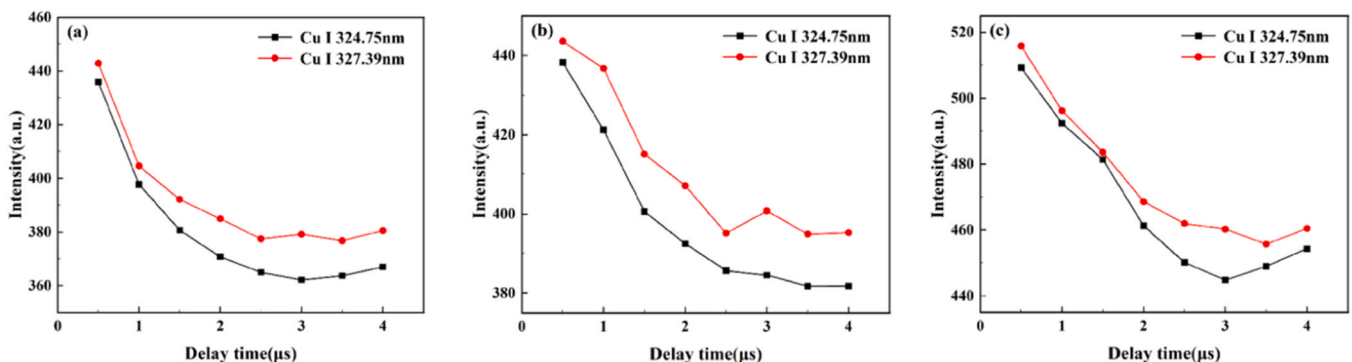


Fig. 4. The change in intensity of Cu spectra with different delay times. (a) *Fritillaria*. (b) *Ferula*. (c) *Cistanche*.

features, presented as a bar plot. To identify the optimal feature subset, the calculated feature importance values were sorted in ascending order, and a SBS strategy was employed for feature selection. The optimal features selected through this process were Mg (280.27 nm), Ca (442.54, 558.87, 612.22, 643.91, 646.25, 649.37, 671.77 nm) and Na (588.99, 589.59 nm). From Fig. 6, it can be observed that the characteristic spectral lines of Cu, Sr, and Si, which exhibit relatively high importance, were not selected by the SBS algorithm. This phenomenon can be attributed to the fact that the feature importance calculated by XGBoost may be influenced by certain features with strong correlations. Even if a feature demonstrates high individual importance, it might be highly correlated with other features, enabling the model to make accurate predictions without relying on it. Consequently, during the SBS process, when optimizing model performance by removing one feature at a time, it is possible to identify that a feature with high importance may have redundant contributions with other features, leading to its exclusion.

To validate the superior classification performance of the proposed XGBoost-MLP model, comparative analyses were conducted against conventional machine learning algorithms, including MLP, XGBoost, and RF using three distinct TCMM species: *Fritillaria*, *Ferula*, and *Cistanche*. As illustrated in Fig. 7, despite the marked spectral differences among these three TCMM categories, conventional algorithms failed to achieve perfect classification accuracy. Panel (a) displays the classification results of the MLP model constructed with features selected by the SBS algorithm, which attained 100 % accuracy on the test sets through 10-fold cross-validation. In contrast, panels (b)–(c) present the confusion matrices of the MLP, XGBoost, and RF models, with corresponding test accuracies of only 99 %, 97 %, and 98 %, respectively, underscoring the limitations of traditional methods even with highly distinguishable samples.

To comprehensively evaluate the effectiveness of our approach, multiple performance metrics, including recall, precision, and F1-score, were employed, as summarized in Table 1. The results demonstrate that the XGBoost-MLP model outperformed all other models across every metric. The experimental findings indicate that XGBoost-driven feature selection effectively eliminates redundant spectral lines, resulting in a

Table 1
The classification results of SSA-XGBoost-MLP, MLP, XGBoost, and RF modeling for three types of traditional Chinese medicinal materials.

Class	Classification algorithm	0(%)	1(%)	2(%)	Average (%)
Precision rate	XGBoost-MLP	100 %	100 %	100 %	100 %
	MLP	100 %	97 %	100 %	99 %
	XGBoost	94 %	97 %	100 %	97 %
	RF	95 %	100 %	100 %	98.3 %
Recall rate	XGBoost-MLP	100 %	100 %	100 %	100 %
	MLP	100 %	100 %	96 %	98.6 %
	XGBoost	97 %	94 %	100 %	97 %
	RF	100 %	94 %	100 %	98 %
F1-score	XGBoost-MLP	100 %	100 %	100 %	100 %
	MLP	100 %	98 %	98 %	98.6 %
	XGBoost	96 %	95 %	100 %	97 %
	RF	97 %	97 %	100 %	98 %

streamlined model architecture that not only accelerates training convergence but also significantly enhances overall efficiency and accuracy. Critically, this successful identification of herb-specific elemental lines (e.g., Mg, Ca, Na) validates XGBoost’s capacity to extract physically meaningful and task-relevant spectral features from complex LIBS data—a capability that directly underpins its subsequent application in the quantitative analysis of trace heavy metals in Section 4.3.

4.3. Quantitative modelling

4.3.1. Evaluation of regression models for quantitative analysis

In this study, we conducted a comparative evaluation of predictive performance across various concentration levels using our proposed hybrid XGBoost-MLP model and conventional machine learning approaches for quantitative analysis of heavy metals in Chinese herbal medicines. Conventional machine learning models (e.g., SVM, MLP, RF) necessitated principal component analysis (PCA) for dimensionality reduction due to the high-dimensional and nonlinear nature of LIBS spectra. While PCA improved computational efficiency through

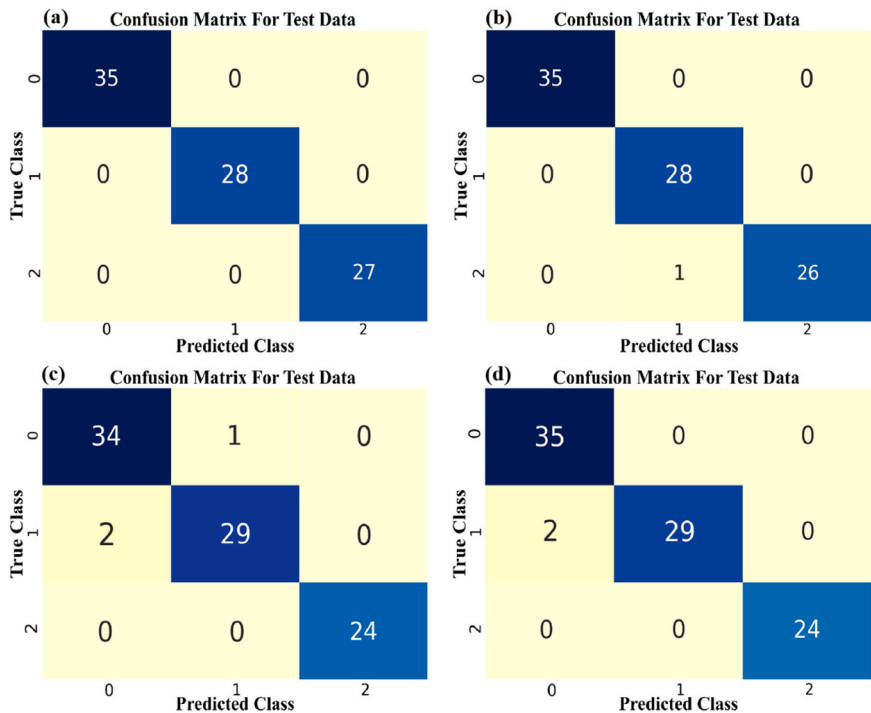


Fig. 7. Classification results of different models on the test set. (a) XGBoost-MLP confusion matrix. (b) MLP confusion matrix. (c) XGBoost confusion matrix. (d) RF confusion matrix.

dimensionality reduction, its global linear transformation mechanism proved inadequate in selectively filtering localized noise associated with matrix effects. In contrast, our hybrid XGBoost-MLP model eliminated PCA dependency through its intrinsic feature selection mechanism. The XGBoost algorithm autonomously identified concentration-critical features via Gain-based importance ranking. Notably, the two Cu characteristic lines (324.75 nm and 327.39 nm) accounted for 98.97 % of cumulative Gain importance, with the top five features collectively contributing 99.99 %. This targeted feature selection enabled precise concentration signal extraction while effectively mitigating interference from matrix-induced noise.

The predictive performance of the proposed XGBoost-MLP model was quantitatively compared against three conventional PCA-based models (PCA-SVM, PCA-MLP, and PCA-RF) through the analysis of the predicted versus actual value plots (Supplementary Fig. S1). Notably, the XGBoost-MLP model achieved the optimal performance on the test set, with a regression equation of $y = 0.952x + 3.21$ ($R^2 = 0.987$), where y is the predicted concentration and x is the actual concentration. This nearly perfect unit slope and high R^2 value indicate its superior accuracy and minimal systematic error. Comprehensive evaluation metrics in Table 2 and Fig. 8 confirmed the hybrid model's optimal performance, particularly excelling in generalization capability and prediction accuracy. While conventional models delivered satisfactory results, the hybrid approach exhibited marked overall advantages.

This study demonstrates that preserving critical spectral features through advanced feature selection significantly enhances prediction accuracy compared to PCA-based dimensionality reduction methods. The proposed XGBoost-MLP framework provides a robust and efficient solution for quantitative heavy metal analysis in Chinese herbal medicines, offering substantial improvements in analytical reliability and operational efficiency.

4.3.2. Comparative analysis of optimization algorithms

To identify the most effective optimization algorithm for hyperparameter tuning, the performance of the SSA was evaluated against two other widely-used metaheuristic algorithms, PSO and the GA. Fig. 9 illustrates the predictive accuracy of the MLP models optimized by these three algorithms on the test set, as measured by R^2 , RMSE, and MAE. The results demonstrate that the SSA-optimized model achieved superior predictive performance, yielding the highest R^2 value along with the lowest RMSE and MAE values. This promising outcome motivated the selection of SSA as the baseline for further development.

However, noticeable fluctuations in the optimization results were observed during the evaluation. Given that stability and reliability are critical requirements for practical applications, the SSA algorithm was subjected to 20 independent runs. The results confirmed its inherent instability. To address this limitation, the proposed MA-SSA was evaluated under identical conditions. Table 3 summarizes the mean and standard deviation of the evaluation metrics for both SSA and MA-SSA across these 20 runs, providing a statistical comparison of their robustness.

As indicated by the substantially smaller standard deviations in Table 3, the proposed MA-SSA algorithm exhibits a remarkable

Table 2

The result of cross-validation and prediction by using different algorithm models.

Algorithm	Sample set	RMSE($\mu\text{g/g}$)	R^2	MRE (%)	MAE($\mu\text{g/g}$)
XGBoost-MLP	Training	7.82	0.982	6.97 %	5.51
	Test	7.17	0.987	7.18 %	5.63
PCA-SVM	Training	6.75	0.986	3.80 %	3.98
	Test	12.63	0.958	10.47 %	7.41
PCA-MLP	Training	10.05	0.970	9.32 %	7.68
	Test	11.18	0.967	9.54 %	7.24
PCA-RF	Training	5.72	0.989	4.32 %	3.62
	Test	15.33	0.939	13.51 %	9.38

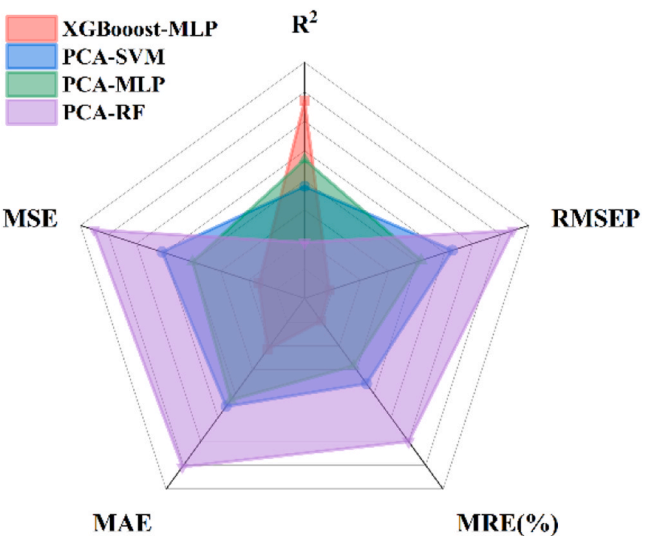


Fig. 8. Comparison of Model Performance Using Different Algorithms Based on Evaluation Metrics.

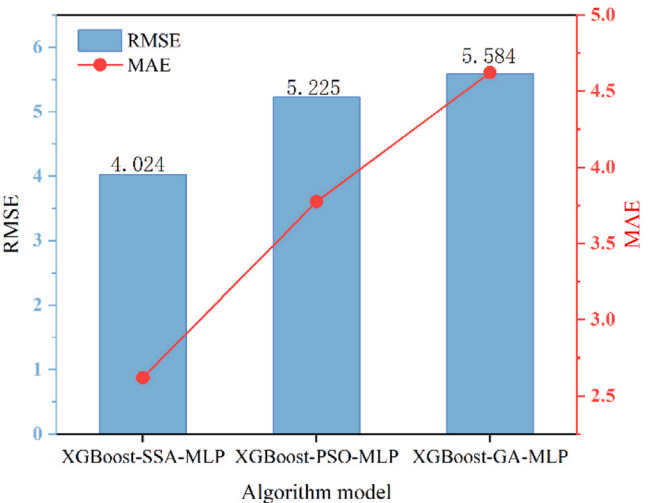


Fig. 9. Performance comparison of MLP models optimized by different algorithms (SSA, PSO, and GA).

Table 3
Stability comparison between SSA and MA-SSA based on 20 independent runs.

Optimization Algorithm	R^2 (Mean $\pm$ SD)	RMSE (Mean $\pm$ SD) ($\mu\text{g/g}$)	MAE (Mean $\pm$ SD) ($\mu\text{g/g}$)
XGBoost-SSA-MLP	0.994 $\pm$ 0.0030	4.62 $\pm$ 1.41	3.19 $\pm$ 1.33
XGBoost-MA-SSA-MLP	0.995 $\pm$ 0.0017	4.43 $\pm$ 0.73	3.18 $\pm$ 0.71

enhancement in stability compared to the standard SSA. Although both algorithms are capable of identifying high-performance solutions, MA-SSA achieves this with consistently greater reliability. The reduction in performance variability is attributed directly to the core improvements integrated into MA-SSA, which effectively mitigate premature convergence and refine the search process.

In summary, the proposed MA-SSA algorithm not only retains the high performance of the standard SSA but also delivers exceptional robustness, making it well-suited for reliable and consistent application in quantitative spectral analysis.

5. Conclusion

This study presents an integrated XGBoost-MLP framework that significantly enhances the classification and quantitative analysis of heavy metals in TCMM using LIBS spectroscopy. The method achieved 100 % classification accuracy among three medicinal species and high predictive accuracy for copper content ($R^2 = 0.987$) by effectively utilizing XGBoost's robust feature selection capability. Unlike PCA-based approaches, the proposed model preserves critical spectral features, reduces redundancy, and mitigates matrix interference. Furthermore, the MA-SSA-optimized MLP demonstrated enhanced stability and consistency over standard SSA, PSO, and GA in regression tasks, with significantly lower standard deviations across multiple runs. This work provides a reliable, efficient solution for heavy metal detection in TCMM, with promising applications in pharmaceutical quality control and environmental monitoring.

CRedit authorship contribution statement

Zhongxin Zhang: Data curation. **Reyihan Tudi:** Data curation. **Bumaliya Abulimiti:** Writing – review & editing, Supervision, Data curation. **Xiang Xiang Mei:** Writing – review & editing, Supervision, Data curation. **Abulimiti Yasen:** Writing – original draft, Visualization, Software, Project administration, Methodology, Data curation, Writing – review & editing. **Zhou Yu xia:** Resources, Methodology. **Wenhan Gao:** Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jpba.2025.117296](https://doi.org/10.1016/j.jpba.2025.117296).

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